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Reducing spatial coherence via dynamic scattering media enables aberration and speckle suppression in optical imaging

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Abstract

High-resolution optical imaging is limited by speckle noise and optical aberrations. Speckle, arising from interference of scattered fields, obscures fine detail and reduces contrast, while aberrations introduce spatially varying blur that is difficult to correct post hoc without additional complex instrumentation or postprocessing. To address these limitations, we show that placing a static sample in a dynamic scattering medium (DSM) produces a sequence of slightly different angular and phase-diverse views of the object. Averaging these views reduces the effective spatial coherence, suppressing speckle and reducing sensitivity to low-order aberrations and static phase-screen distortions with a small, predictable blur. A simple imaging model captures this behavior. Small random angular deviations and fine phase fluctuations introduced by moving scatterers average out as noise while contributing to blur. We further demonstrate that complex (field) averaging, after phase stabilization, provides higher contrast than intensity averaging. Simulations quantify the trade-off between speckle suppression and blur and agree with experiments in which we image a USAF resolution chart beneath a 250-μm mouse brain slice. Our method requires only repeated, independent views and is demonstrated in coherence-gated interferometric (Fourier-domain OCT) imaging. The mechanism is in principle applicable to other coherent imaging modalities capable of acquiring repeated, independent views. These findings suggest a pathway toward dynamic-diffuser-immersed objectives – optical elements that intrinsically suppress speckle and reduce sensitivity to low-order aberrations through controlled spatial-coherence modulation.

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Introduction

High-resolution imaging systems, including optical coherence tomography (OCT) [1], digital holography [2], interferometric microscopy [3, 4], confocal and wide-field microscopes [5], and beam-scanning platforms, are fundamentally limited by speckle noise and optical aberrations [4, 6].

Reducing the effective spatial coherence of illumination is a well-established strategy for suppressing speckle [7]. We use the term dynamic scattering medium (DSM, introduced in the abstract) to refer to a thin layer containing moving scatterers that generates time-varying speckle fields. Mechanical diffusers, rotating phase plates, deformable mirrors, and spatial light modulators can generate rapidly varying speckle fields whose incoherent averaging improves image contrast [8-13]. However, these approaches may introduce angular diversity that degrades resolution, are limited by modulation speed, and require careful alignment and registration - issues that are especially problematic in biological imaging, where bulk motion and sample deformation are common.

Optical aberrations impose an additional major limitation on imaging performance. Traditionally, aberrations are minimized through sophisticated lens design or corrected dynamically using adaptive optics (AO) [14, 15]. AO, originally developed for astronomy [14], is widely used to compensate sample- and system-induced wavefront errors in applications such as retinal imaging [16, 17], super-resolution microscopy [18] and OCT [19]. AO can improve resolution and sensitivity, but AO systems remain mechanically and computationally complex, requiring deformable mirrors, wavefront sensors or sensorless optimization loops, and continual recalibration [15]. These demands make AO expensive, alignment-sensitive, and difficult to deploy in compact or long-term-stable instruments, particularly in scattering biological tissues where the wavefront varies rapidly and unpredictably. A few recent studies suggest that reducing spatial coherence can modestly decrease sensitivity to certain aberrations [20], but the effect has been limited and remains underexplored.

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Beyond low-order aberrations, strongly scattering samples introduce high-spatial-frequency distortions that further scramble the wavefront, producing speckle and limiting penetration depth and image fidelity. Wavefront-shaping techniques [21], including transmission-matrix estimation [22], iterative optimization, and optical phase conjugation, can recover or enhance fields behind complex media and enable focusing through opaque layers [23]. Nevertheless, they often require high-speed modulation, and long optimization times, complicating integration into routine biological imaging where tissues drift and deform.

An alternative strategy for overcoming scattering is to isolate the small fraction of ballistic and quasi-ballistic photons that retain directional information after propagating through turbid media [24-26]. Time gating [27-29], coherence gating [30], and confocal rejection [31] selectively detect these minimally scattered components, enabling high-resolution imaging in highly scattering tissues. OCT [1] is a prominent example: by using low-coherence interferometry to reject multiply scattered light, it achieves micrometer-scale axial sectioning and penetration depths of 1–2 mm [30] in biological samples. Yet despite its powerful gating capabilities, OCT and similar systems remain limited by speckle originating from residual forward-scattered fields [4, 6], as well as by sample- and system-induced aberrations. These factors degrade contrast, obscure fine structural detail, and limit transverse resolution, even in carefully engineered or AO-augmented systems.

Temporal fluctuations within the sample can provide functional contrast. Dynamic light scattering [32] and diffuse correlation spectroscopy [33, 34] use motion-induced decorrelation to probe flow and microscopic dynamics, while speckle-contrast imaging [35] maps blood flow by analyzing temporal speckle variations. In OCT, temporal decorrelation reveals cellular motility [36], axial motion, and even stimulus-evoked optoretinographic responses [37]. These temporally varying speckles decorrelate across acquisitions, so averaging can suppress motion-related fluctuations and reveal the static structure more clearly. In static samples, however, the absence of intrinsic motion means that consecutive acquisitions remain highly correlated, preserving both speckle and aberration artefacts unless external diversity is introduced. Controlled, externally driven diversity is therefore required when imaging stationary structures.

These two perspectives, using ballistic or quasi-ballistic photons for structural imaging, and exploiting dynamic scattering for flow or motility contrast, are typically regarded as distinct and mutually exclusive. Ballistic imaging preserves resolution by rejecting multiple scattering, whereas conventional dynamic-scattering approaches operate in the diffuse regime, where multiply scattered photons dominate and encode motion. Consequently, dynamic media are usually assumed to degrade high-resolution imaging, since additional scattering is expected to blur fine detail and reduce contrast. We therefore ask whether a weak, controlled dynamic layer can supply sufficient angular and phase diversity to suppress speckle and reduce aberration sensitivity while keeping diffuse blur small.

Dynamic scattering media provide controlled temporal diversity for coherence modulation. Fluids containing Brownian particles [32], gently flowing suspensions, dynamic phantoms, or perturbed multimode fibers naturally generate time-varying speckle fields. As the scatterer distribution evolves, the resulting speckle patterns decorrelate, enabling incoherent averaging of sequential measurements while preserving high-resolution information. Prior work in correlation-gated [38] and time-varying speckle imaging has shown that such dynamics can be used to modulate coherence or isolate minimally scattered components.

We test whether a weak dynamic scattering layer can suppress speckle and reduce aberration sensitivity by lowering the effective spatial coherence [38-42]. As the speckle field decorrelates, sequential complex wavefronts sample slightly different angular and phase configurations; averaging then suppresses coherent artefacts at the cost of a small, predictable blur. We quantify this trade-off in simulations and validate it experimentally in OCT. Dynamic scattering media provide a passive means of tailoring spatial coherence, enabling speckle suppression and reduced aberration sensitivity without mechanical diffusers, spatial light modulators, adaptive optics, or complex phase-control hardware.

We implemented DSM coherence modulation in a Fourier-domain OCT system. Coherence gating selects ballistic and quasi-ballistic photons, and the complex-valued interferometric signal enables direct comparison of coherent (field) and incoherent (magnitude) averaging. The micrometer-scale axial resolution and phase sensitivity allow quantitative assessment of speckle suppression and aberration mitigation.

Speckle reduction via decorrelation and averaging is a well-established strategy in coherent imaging. Approaches such as moving or rotating diffusers at the pupil, speckle-modulating OCT (SM-OCT), partially spatially coherent illumination (e.g., LED or multimode sources), and Spatiotemporal Optical Coherence (STOC) manipulation all exploit this principle. Accordingly, we do not claim novelty in coherence reduction or averaging per se.

Instead, our contribution is threefold. First, we introduce a passive, sample-adjacent dynamic scattering medium (DSM) that acts as an in-situ coherence modulator, eliminating the need for moving optics, spatial light modulators, or active wavefront control. Second, we develop an analytic imaging model that explicitly separates two effects: (i) effective PSF broadening governed by the angular tilt variance σ_θ, and (ii) contrast loss and pedestal formation determined by dynamic phase variance σ_{dyn}, the number of realizations N, and any static phase component. This separation yields simple and actionable design rules. Third, leveraging the interferometric nature of OCT, we directly compare magnitude and complex (field) averaging, demonstrating that complex averaging suppresses the incoherent pedestal that persists under intensity-only averaging.

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A related line of work on imaging through dynamic turbid media (e.g., [43]) exploits angular decorrelation of ballistic light to recover images through static aberrations. In contrast, our approach combines OCT coherence gating with averaging across DSM realizations and does not rely on a computational image-recovery step. Concretely, we (i) formulate a time-dependent pupil model in which the DSM is represented as a combination of small random tilts and higher-order random phase perturbations; (ii) derive closed-form expressions for the effective PSF and the residual pedestal under both magnitude and complex averaging; (iii) validate the predicted Gaussian blur kernel against numerically estimated effective PSFs obtained from averaged simulated images; (iv) demonstrate experimentally, in Fourier-domain OCT, that DSM-enabled averaging suppresses speckle while reducing sensitivity to low-order aberrations and static phase-screen distortions (e.g., Scotch tape and a 250 μm mouse brain slice), at the cost of a modest and predictable resolution loss; and (v) quantify the required acquisition delay directly from the measured DSM decorrelation time.

As illustrated in Fig. 1, the incident beam passes through a thin dynamic scattering layer containing moving microscopic particles (e.g., Brownian scatterers). Each particle perturbs the optical path by introducing small, random changes in phase and scattering angle. As the scatterers evolve over time, the illumination arriving at the underlying static object becomes slightly different for each acquisition. Importantly, these fluctuations are weak: most of the light remains near-ballistic, so the spatial resolution is preserved, but the wavefront is no longer identical from one measurement to the next.

Each illumination realization therefore produces a distinct speckle field upon reflection from the static object. In OCT, this results in time-varying speckle patterns in successive complex-valued interferograms. Because these speckle patterns are uncorrelated, averaging multiple acquisitions reduces the coherent artefacts associated with stable speckle and low-order aberrations.

If the complex-valued OCT signal is phase-stabilized, coherent (complex) averaging [13] combines repeated wavefronts constructively for the static sample while cancelling the random speckle fluctuations, providing the greatest improvement in contrast. Without explicit phase alignment, incoherent (magnitude) averaging still reduces speckle but is less effective at recovering low-contrast features. In both cases, the evolving scattering layer supplies the necessary angular and phase diversity for speckle suppression without introducing macroscopic motion or large-angle blur.

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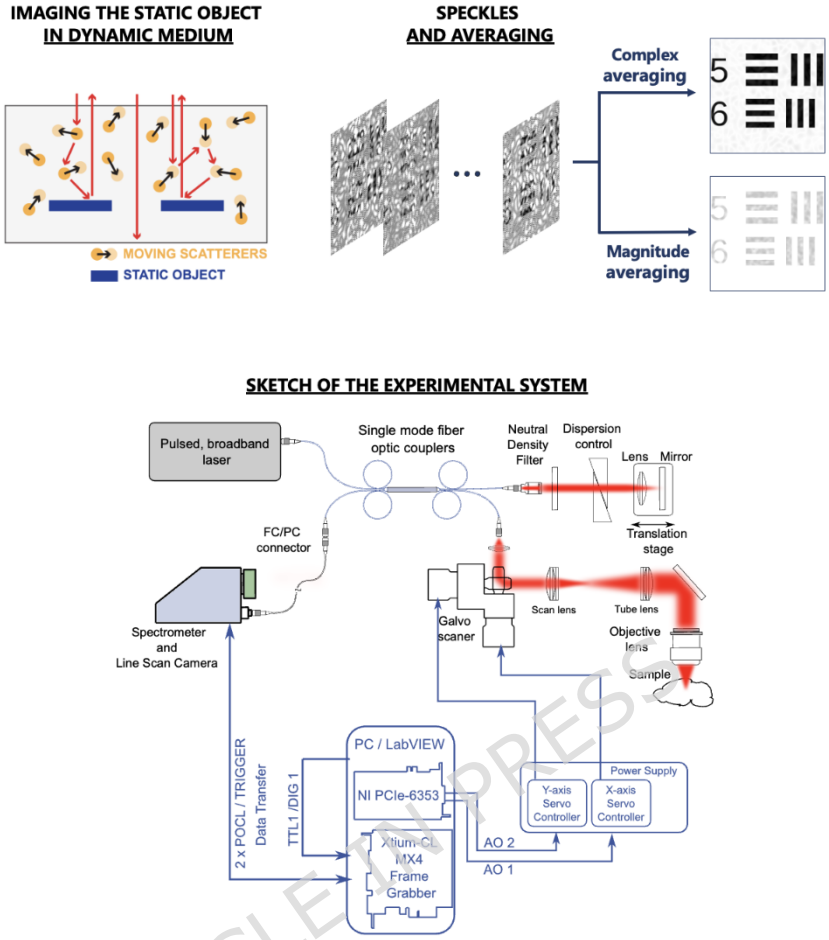


Fig. 1. Principle of dynamic-scattering–induced coherence modulation (top) and the sketch of experimental system (bottom). Light passes through a dynamic scattering layer before and after interacting with the static sample. Moving microscopic scatterers introduce small, time-varying phase shifts and angular deviations, producing a different illumination wavefront for each acquisition. These fluctuations generate independent speckle realizations in the OCT signal. By averaging multiple decorrelated measurements, either coherently (after phase stabilization) or incoherently (magnitude averaging), the static structural signal is reinforced while random speckle and aberration-related artefacts are suppressed.

Theoretical Background

Throughout this work we use the following conventions. Aberration refers to a pupil-plane wavefront phase error (e.g., defocus, astigmatism, Zernike-decomposable wavefronts). Blur refers to its image-plane consequence: a broadened point-spread function (PSF) or reduced modulation transfer function. Distortion and static phase-screen effects refer to spatially varying phase in or near the sample plane (e.g., Scotch tape, brain slice) that produces coherent artefacts such as speckle-like structures rather than a pure defocus blur. Consistent with these definitions, we describe our method as reducing sensitivity to low-order aberrations and suppressing coherent artefacts introduced by static phase-screen distortions, rather than correcting aberrations.

We model the dynamic scattering medium (DSM) as a time-dependent pupil that adds small random angular deviations (tilts) and higher-order random phase to any static, low-order aberration in the optics. The object-plane field is written from the outset as a ballistic term U_0 plus a DSM-induced dynamic term

$$U(r,t) = U_0(r) + U_{dyn}(r,t) \quad (1)$$

with r the image-plane coordinate. The dynamic component U_{dyn} is the zero-mean, multiply-scattered field generated by the moving particles. Its second moment produces an incoherent background under intensity averaging (the dynamic pedestal defined below).

With pupil coordinates $\rho = (x,y)$ normalized to the pupil radius, the complex pupil at time t is:

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$$H(\rho,t) = A(\rho)\exp\{i[\Phi_{ab}(\rho) + 2\pi(\theta_x(t)x + \theta_y(t)y) + \Phi_{dyn}(\rho,t)]\}, \quad (2)$$

where $A(\rho)$ is the aperture, $\Phi_{ab}(\rho)$ encodes static low-order aberrations (e.g. defocus), θ_x,θ_y are zero-mean random tilts (angular diversity), and $\Phi_{dyn}(\rho, t)$ is a zero-mean higher-order random phase due to particle motion. The coherent impulse response and intensity PSF follow as

$$h(r,t) = F^{-1}\{H(\rho,t)\}, \quad PSF(r,t) = |h(r,t)|^2, \quad (3)$$

with image-plane coordinate r. We use unit-energy normalization $\sum_h |h(r,t)|^2 = 1$, so the optical transfer function satisfies $OTF(0,t) = 1$.

Equations (1)-(3) treat the DSM as a thin, weakly scattering layer acting as a single-pass random phase screen in the small-angle regime. For our 3.2% milk DSM, this is justified because the layer is thin (of the order of one millimeter) and at centre wavelength of ~800 nm milk is scattering-dominated ($\mu_s \gg \mu_a$) but the reduced scattering coefficient after dilution is modest (of order 1 per mm) so the ballistic component is preserved within the OCT dynamic range [44]. We therefore treat Eq. (2) as a phenomenological model that captures the measured angular and phase diversity without requiring a first-principles treatment of the milk microphysics. When the DSM becomes optically thick, the ballistic fraction decays exponentially and the effective angular spread grows. The blur law (Eq. 9) still applies, but the resolution penalty grows and the method eventually fails when the coherent peak of the PSF is destroyed.

A thin static phase screen near the object produces static speckle in a single frame. We represent it by an object-plane phase $\Phi_{stat}(r)$ with variance σ^2_{stat} and correlation length l_{stat}. So, the ballistic field becomes

$$U_{stat}(r) = A_0(r)e^{i[\varphi_0(r)+\Phi_{stat}(r)]}. \quad (4)$$

where A_0 and φ_0 are the object's amplitude and phase. For the k-th realization (tilt and dynamic phase),

$$V_k(r) = [U_{stat} * h_k](r), \quad I_k(r) = |V_k(r)|^2, \quad (5)$$

with $h_k = h(r,t_k)$ and $*$ denoting convolution. If $\sigma_{stat} \gtrsim \pi$ (over a resolution cell), the single-frame intensity I_k exhibits fully developed static speckle [6]. For $\sigma_{stat} \ll 1$ rad, a coherent component survives with an attenuation factor $e^{-\sigma^2_{stat}/2}$ and a weak speckle halo.

Because DSM-induced tilts and dynamic phase re-sample how Φ_{stat} interferes, the static speckle decorrelates across realizations. After phase stabilization, complex (field averaging) [15] yields

$$\langle V_k(r)\rangle \approx U_0 * h_0 * g_{\sigma,\theta} \times m_{stat} \quad (6)$$

where $U_0 = A_0e^{i\phi_0}$, $h_0 = F^{-1}\{Ae^{i\Phi_{ab}}\}$ is the static coherent core, g_{σ_θ} is a Gaussian blur induced by the tilt variance $\sigma^2_\theta = \langle\theta^2_x + \theta^2_y\rangle$, and $m_{stat} \in (0,1]$ is the coherent reduction factor from the static phase (for weak screens $m_{stat} \approx e^{-\sigma^2_{stat}/2}$). Thus, complex averaging suppresses the random speckle term and leaves an effective blur governed by the tilt variance.

By contrast, magnitude (intensity) averaging retains the second-moment contributions that do not cancel in the mean:

$$\langle|V_k(r)|^2\rangle \approx |U_0 * h_0|^2 * g_{\sigma_\theta} + \langle|U_{stat} * h_k|^2\rangle_{incoh} + \langle|U_{dyn} * h_k|^2\rangle. \quad (7)$$

Here the static halo (the second term) is a slowly varying background set by l_{stat} and the broadened core $|h_0|^2 * g_{\sigma_\theta}$. The dynamic pedestal (the last term) is the strictly non-negative, incoherent background from DSM-induced multiply scattered light U_{dyn}. Both terms persist in the mean (their variance shrinks as $1/\sqrt{N}$ with the number of averages N). In practice, the pedestal acts as a non-negative additive background that reduces Michelson contrast:

$$MC = \frac{I_{max}-I_{min}}{I_{max}+I_{min}} \quad \rightarrow \quad \frac{I_{max}-I_{min}}{I_{max}+I_{min}+2C}, \quad C \geq 0. \quad (8)$$

Complex averaging removes the pedestal and suppresses the granular part of the halo, while the magnitude averaging does not.

In both averaging modes the effective PSF is the coherent core convolved with g_{σ_θ}. If this kernel is Gaussian with image-plane standard deviation σ_g, the transverse resolution obeys the blur law

$$FWHM^2_{eff} \approx FWHM^2_0 + 4\ln 2\sigma^2_g, \quad \sigma_g \propto f\sigma_\theta, \quad (9)$$

For a Gaussian, $FWHM = 2\sqrt{2\ln 2}\sigma$. Variances add under convolution, giving the $8\ln 2\sigma^2_g$ factor in $FWHM^2$

where f is the image-plane scaling factor that converts pupil-plane tilt to lateral blur for a given system (it absorbs focal length, magnification, wavelength, and NA). The static phase reduces coherent throughput via m_{static} and adds a local halo in magnitude averages, but it does not change the functional form of the blur law.

Independence between realizations is ensured by choosing the inter-volume delay Δt above the DSM decorrelation time. For Brownian motion [23], $\langle\Delta r^2\rangle = 4D\Delta t$, requiring RMS displacement ~$\lambda/2$ gives

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$\Delta t \approx \frac{\lambda^2}{16D}$. (10)

This sets a lower bound for independent views. In practice, Δt can be chosen from the measured decay of the field autocorrelation $g_1(\tau)$. With this choice, successive realizations are effectively independent, complex averaging removes the dynamic pedestal (since $\langle U_{dyn} \rangle = 0$), and magnitude averaging exhibits the expected $1/\sqrt{N}$ speckle-variance decay [6] while retaining the mean offset. In experiments we used a longer delay guaranteeing decorrelation.

In summary, a thin static phase screen explains static speckle in a single frame. DSM-induced tilt and random phase decorrelate that speckle across views. After averaging, complex images show a blurred but high-contrast coherent reconstruction (distortions suppressed, no pedestal), whereas magnitude images show the same blur plus a residual halo/pedestal that depresses Michelson contrast.

Methods and Results

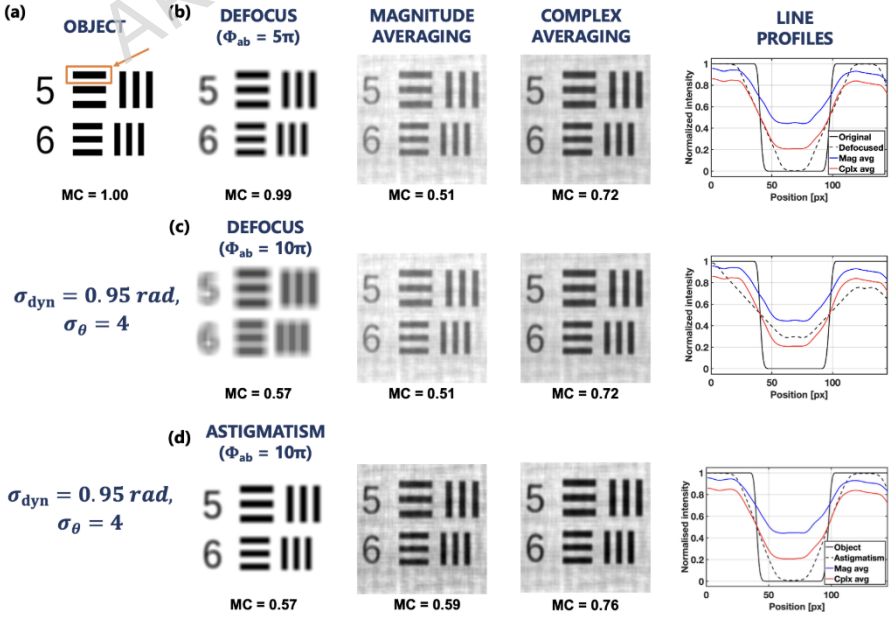
Numerical Simulations

We simulated DSM-induced diversity in the presence of a pure defocus aberration. Starting from a baseline image formed with a static pupil defocus Φ_{ab} , we generated N=100 independent views by applying, per view, a random tilt (rms σ_θ) and a higher-order zero-mean dynamic phase (rms σ_{dyn}). We then compared magnitude (intensity) averaging with complex (field) averaging (Fig. 2).

With $\sigma_{dyn} = 0.95$ rad and $\sigma_\theta = 4$ (normalized pupil units), both complex and magnitude averages suppress speckle and defocus ripples, and both blur edges according to the blur law (same σ_θ leads to similar transverse resolution, Δx). The primary difference is the additive pedestal: magnitude averaging retains a non-negative background term that lowers MC (≈ 0.51), whereas complex averaging suppresses the pedestal. Increasing defocus from 5π to 10π lowers the baseline contrast, but the post-averaging contrast separation remains because it is dominated by the pedestal, not defocus amplitude.

The MC values reported here are computed on the grand average over all N = 100 realizations. A detailed uncertainty analysis based on 20 random subset averages of size N are provided in Section S6 of the Supplement.

To assess generality beyond defocus, we repeated the analysis for a non-symmetric aberration (astigmatism, $\Phi_{ab} = 10\pi$). The same trends are observed: DSM-induced averaging mitigates aberration artifacts and enforces a comparable blur set by σ_θ, while complex averaging consistently preserves higher contrast by suppressing the dynamic pedestal. This indicates that the mechanism is not specific to defocus but extends to broader classes of low-order aberrations



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Fig. 2. DSM-induced averaging suppresses defocus artifacts. (a) Object (boxed region indicates the line-profile location). (b) Baseline defocus and DSM-averaged reconstructions for Φ_{ab} = 5π: defocus only (MC = 0.99), magnitude averaging (MC = 0.51), complex averaging (MC = 0.72), and the corresponding line profiles (right). (c) Same as (b) for stronger defocus Φ_{ab} = 10π: defocus only (MC = 0.57), magnitude (MC = 0.51), complex (MC = 0.72). (d) Astigmatism (Φ_{ab} = 10π): aberrated image (MC = 0.57), magnitude averaging (MC = 0.59), and complex averaging (MC = 0.76), with corresponding line profiles (right). In all cases, DSM-induced averaging (N=100 realizations) reduces sensitivity to aberrations and produces an effective blur governed by the same angular variance σ_θ. Both magnitude and complex averaging yield comparable edge broadening, consistent with the predicted blur law, while complex averaging preserves higher contrast by suppressing the incoherent (dynamic) pedestal. This is reflected in the line profiles as a raised valley for magnitude averaging (contrast loss) versus a deeper valley for complex averaging.

To decouple contrast from resolution, we fixed $\sigma_{\theta} = 1$ and varied the dynamic phase depth σ_{dyn} (Fig. 3). Decreasing σ_{dyn} from 0.95 rad to 0.5 rad reduces the dynamic pedestal and increases Michelson contrast (MC ≈0.97 for complex averaging and ≈0.79 for magnitude averaging), while edge widths remain nearly unchanged because the blur kernel is set by σ_θ. These trends follow Eqs. (7–8): σ_θ controls the effective PSF width, whereas σ_{dyn} and N primarily control the pedestal and residual speckle variance.

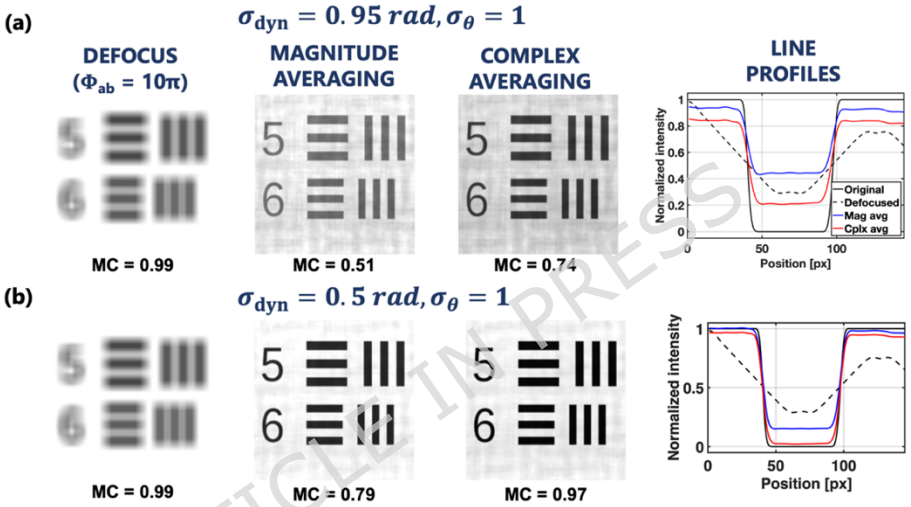


Fig. 3. Contrast versus dynamic phase depth at fixed blur. With the angular diversity fixed (σ_θ = 1), the blur remains approximately constant, isolating contrast changes. (a) With a relatively strong dynamic phase (σ_{dyn} = 0.95 rad) and defocus Φ_{ab} = 10π, the magnitude average carries a noticeable pedestal and yields shallow valleys (MC ≈ 0.51), while the complex average, free of that pedestal, recovers a visibly deeper profile (MC ≈ 0.74). (b) When we reduce the dynamic phase to σ_{dyn} = 0.5 rad, both curves sink further between bars and the Michelson contrast rises (to ≈ 0.79 for magnitude and ≈ 0.97 for complex), yet the edge widths remain essentially unchanged because σ_θ has not changed. We used N=100 realizations.

Speckle suppression

We added a time-invariant phase screen Φ_{stat} to the object and acquired N=100 independent views with fixed DSM diversity (σ_{dyn} = 0.95 rad, σ_θ = 1). Figure 4 explores four combinations of the screen's correlation length (l_{stat}) and strength (σ_{stat}): fine/coarse grain (1 vs 25 px) and moderate/strong phase (0.8 vs 2.5 rad).

Fig. 4(b) shows the single image in each case. The speckle contrast (K = σ/⟨I⟩) measured in a fixed bright ROI (orange box) is high (K ≈ 0.69–0.93). After averaging (rows c–d), the results are consistent with the model: both magnitude and complex averaging suppress granular speckle across all conditions, while the overall blur remains similar (fixed σ_θ). Magnitude averaging yields consistently low K (≈ 0.06–0.09) but carries a bright background (the additive pedestal), which visually decreases the contrast of the bars. Complex averaging removes the pedestal and recovers a cleaner coherent image; the background is darker because the coherent throughput is attenuated by the static screen ($m_{stat} \approx e^{-\sigma^2_{stat}/2}$). In the strong-screen regime (σ_{stat} = 2.5 rad), the coherent component is small, lowering the mean ROI intensity; consequently, the reported K can increase numerically (0.09–0.22) despite improved visual

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structure. This is consistent with the model: complex averaging suppresses the pedestal, whereas magnitude averaging retains a residual offset; the edge blur is determined by σ_θ.

Note that K is computed in the same ROI. In the strong-screen cases the lower mean intensity of the complex image inflates K numerically despite the absence of the additive pedestal.

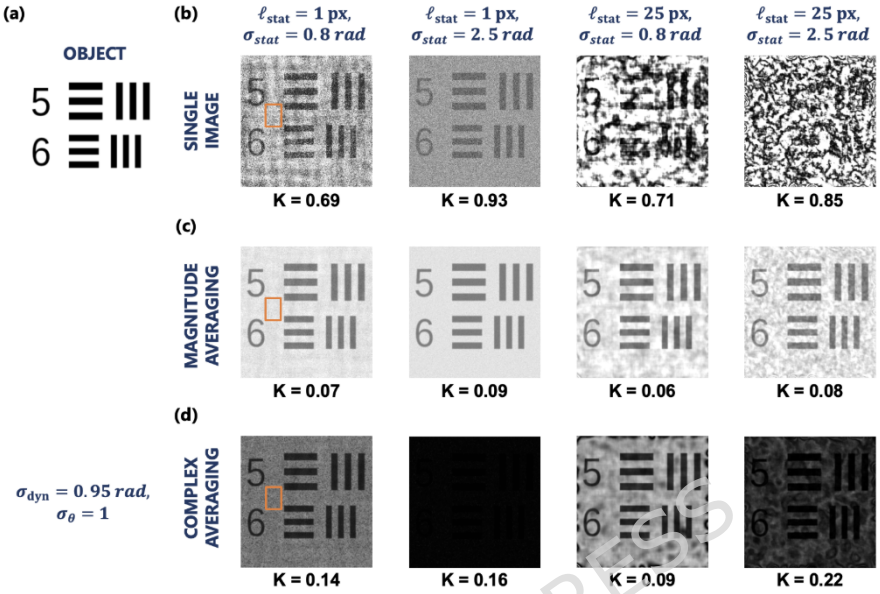


Fig. 4. Static-speckle suppression with DSM averaging. (a) Test object. (b) Single images for four static phase screens for variable l_{stat} and σ_{stat}. Static speckle is strong, which is confirmed by large speckle contrast values (K) between 0.69 to 0.93. (c) Magnitude averaging suppresses the granular pattern but leaves a bright additive background (pedestal). (d) Complex averaging removes the pedestal and reveals a coherent image on a darker background (coherent throughput reduced by η_{stat}). Blur is similar in all columns because σ_θ is fixed. We used N=100 realizations.

Imaging USAF Resolution Chart immersed in dynamic scattering medium

To experimentally demonstrate the above effects, we immersed a USAF resolution test chart (static object U_0) in milk (3.2% fat), which served as the dynamic scattering medium (DSM). Our spectral-domain OCT system utilized a femtosecond laser (Fusion Femtolasers, Austria) with a central wavelength of 795 nm and a full bandwidth of 130 nm, resulting in a measured axial resolution of 2.9 µm (3.9 µm in air). The Michelson interferometer configuration employed a fiber-based beam splitter with a 75/25 splitting ratio. The sample was probed with linearly polarized light. The light reflected from the sample was combined with light reflected from the mirror in the reference arm of the interferometer. The resulting spectral signal, which includes contributions from both the static object and the dynamic scattering caused by the milk, was recorded using a high-speed (250k A-lines per second) 2048-pixel line-scan camera coupled with a commercial spectrometer (Cobra-S 800, Wasatch Photonics, USA).

A 10x microscope objective (Olympus UPlanFL N, numerical aperture 0.3, working distance 10 mm) was used to illuminate the sample. The objective pupil was illuminated with a 6-mm ($1/e^2$) Gaussian beam, resulting in theoretical and experimental lateral resolutions of 4.4 µm and 4.6 µm, respectively. Samples were illuminated with an average optical power of 150 µW. Lateral beam scanning was performed using galvo scanners (Thorlabs Inc., USA). 3D volumes, typically covering an area of approximately 2×2 mm, were acquired with a pixel density of 512×512 along the X and Y axes. The imaging depth, limited by light penetration and multiple scattering, was approximately 800 µm. Custom LabVIEW software managed the scanning protocol and synchronized camera and scanner operations. Raw spectral data underwent standard OCT processing steps, including DC component removal, spectral calibration-based resampling, numerical dispersion compensation, and spectral shaping.

To ensure sufficient decorrelation between consecutive OCT signals, we first estimated the minimum required time delay (Δt) using the Brownian diffusion model, where the mean squared displacement is defined as $\langle \Delta r^2 \rangle = 4D\Delta t$ (with D representing the diffusion coefficient). To achieve adequate spatial decorrelation,

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scatterers must move by at least half the illumination wavelength ($\lambda/2$), resulting in $\Delta t = \lambda^2/(16D)$. For $\lambda = 795$ nm and a diffusion coefficient typical of milk ($D = 1\times10^{-8}$ cm²/s) [24], the calculated minimum Δt is approximately 0.4 seconds.

We used longer delays (60 s) between consecutive acquisitions to ensure complete decorrelation. This increased sensitivity to interferometer drift. To address drift-induced phase instabilities, particularly significant during complex averaging, we applied a two-step phase-correction procedure focused on predefined axial regions corresponding to the USAF resolution chart location. First, spatial phase correction was performed on each sub-volume using Gaussian filtering to smooth out spatial phase variations. Next, temporal phase correction was applied by computing phase differences relative to a stable reference frame derived from spatially corrected data.

We did not measure a continuous decorrelation curve because the 60 s acquisition cadence is much coarser than the expected decorrelation time; we computed g_1 between consecutive complex volumes (the lower-left panel in Fig. 5) and its decay to baseline confirms the statistical independence required by the averaging analysis. The 60 s delay is therefore a deliberate conservative margin above the Brownian estimate of ~0.4 s.

After the two-step phase correction, the residual phase fluctuation measured on stable reference voxels was σ_{phi} ~ 0.2-0.4 rad RMS (equivalent to an optical-path-length fluctuation of approximately 25-50 nm at 795 nm). The Gaussian smoothing scale for the spatial phase correction was chosen to be larger than the measured speckle correlation length but smaller than the target feature size; the smoothing scale used here was chosen at the small end of the trade-off explored in Section S4 (Fig. S4): aggressive smoothing reduces sigma_phi but simultaneously erodes the coherent residual that complex averaging is designed to preserve. The compression of the experimental MC gap (0.84 vs 0.81) relative to the simulated gap (0.72 vs 0.51) is the consequence of operating at a smoothing scale where sigma_phi is acceptably small while the coherent contribution is partly washed out; see Section S4 for the quantitative analysis.

We acquired a series of 100 volumes with these extended delays to ensure robust decorrelation. Initially, the USAF target was imaged alone. Subsequently, for the DSM+OCT experiments, the target was embedded within milk. With the objective above the sample, surface reflections from the milk–air interface were comparable in magnitude to the target backscatter. However, surface reflections could become more problematic when imaging weakly reflecting samples, such as in vivo tissues. To mitigate this, immersing the objective lens directly into the scattering medium can reduce the surface reflections.

After phase stabilization and registration, we compared *en face* averages for magnitude and complex reconstructions (Fig. 5). The measurements reproduce two trends observed in simulation. First, contrast decreases relative to OCT-only because magnitude averages retain a small additive background, while complex averages reduce it. Second, edges broaden according to the blur law (same σ_θ in both modes). Quantitatively, MC was 0.81 (magnitude) vs 0.84 (complex). Edge-spread fits gave $\Delta x \approx 7.7$ µm (magnitude) and 7.5 µm (complex) compared with 4.6 µm (OCT-only), in agreement with a modest σ_θ. The residual MC gap is smaller than in simulation due to residual phase drift and necessary phase smoothing (excessive smoothing reduces the contrast advantage of complex averaging). The autocorrelation g_1 confirms decorrelation between volumes (lower left of Fig. 5). The MC and edge-spread values reported here are computed on the grand average over all N = 100 acquired volumes; bootstrap convergence over 20 random subsets of size N is reported in Section S6 (simulation) and Section S7 (experimental brain-slice dataset).

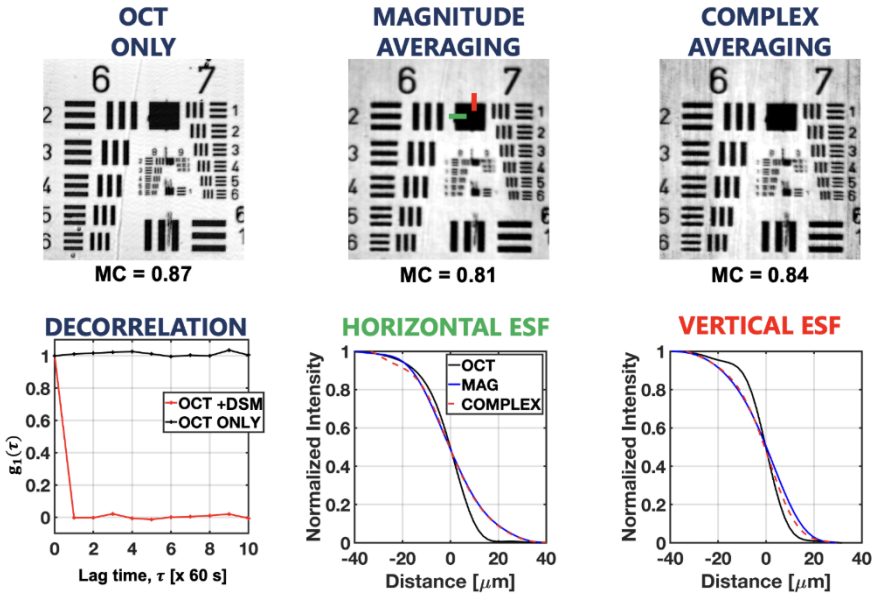


Fig. 5. Comparison of magnitude and complex averaging in DSM-enhanced OCT (N=100 realizations). Averaging broadens the edge-spread function (ESF), increasing Δx relative to OCT-only. The autocorrelation function, g_1, computed between consecutive volumes confirms decorrelation in the OCT+DSM (milk) case. Images and corresponding ESFs are displayed on a linear scale.

Suppressing Distortion

We assessed distortion mitigation by imaging a USAF resolution chart covered with Scotch tape and embedded in milk (Fig. 6). The tape acts as a static phase screen (cf. the simulated Φ_{stat} cases). Averaging suppresses its granular distortions, while the blur is the same trade-off governed by σ_θ. Both magnitude and complex averaging effectively suppressed layer-induced distortions without requiring digital aberration correction. Using a dynamic scattering medium reduces the effective spatial coherence, yielding performance comparable to OCT systems that employ spatially incoherent illumination and therefore show reduced sensitivity to low-order geometrical aberrations [20].

This mechanism is analogous to the Spatiotemporal Optical Coherence (STOC) manipulation method, where time-varying inhomogeneous phase masks modulate incident illumination to reduce spatial coherence [39]. Similarly, in our approach, milk acts as a dynamic medium generating time-varying speckle patterns. However, unlike STOC, our DSM approach introduces spatial blurring, slightly reducing the resolution, as quantified in Fig. 2. Thus, while both methods effectively suppress distortions, STOC maintains spatial resolution, whereas DSM-enhanced OCT involves a trade-off between effective distortion mitigation and spatial resolution. This trade-off can be reduced with complex averaging provided interferometer phase stability is maintained or corrected in post-processing.

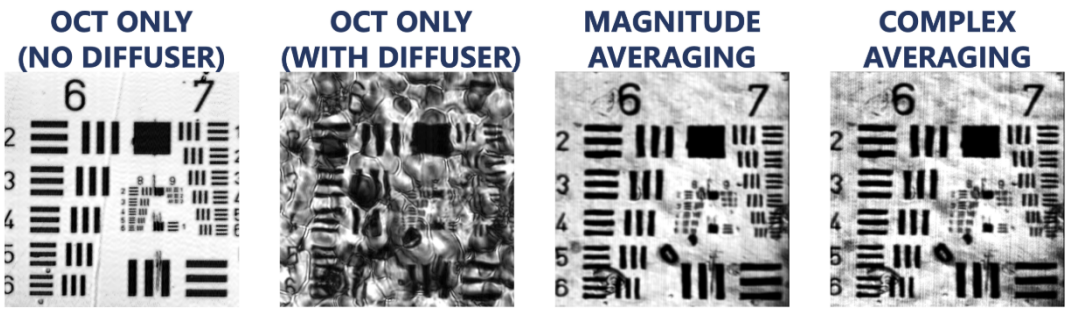


Fig. 6. Distortion suppression using DSM-enhanced OCT (milk layer and N=100). The "with diffuser" case introduces static distortions; magnitude and complex averages are computed over decorrelated DSM realizations, reducing the distortions. Residual distortions in complex averaging are due to phase instability.

Imaging USAF Resolution Chart Hidden by Thin Brain Slice

Finally, we imaged a USAF resolution test chart placed beneath a 250 µm-thick mouse brain tissue slice. Mice were anaesthetised with isoflurane and euthanised by transcardial perfusion. Animals were first perfused with phosphate-buffered saline (PBS) to remove blood, followed by 4% paraformaldehyde (PFA) for fixation. Brains were then dissected and post-fixed overnight in 4% PFA, followed by cryoprotection in 10%, 20% (1 day each) and 30% sucrose (3 days) prior to cryosectioning (Kriostat SLEE MEV/MEV+). All procedures were performed in accordance with Directive 2010/63/EU. The International Centre for Translational Eye Research (ICTER), Institute of Physical Chemistry, Polish Academy of Sciences, is authorised by the Polish Ministry of Science and Higher Education as User no. 16.

The brain layer behaves as a strong, coarse static phase screen with attenuation, consistent with the ($l_{stat} = 25$ px, $\sigma_{stat} \gg 1$) regime in simulation. Embedding the sample in milk and averaging over 100 volumes (magnitude and complex) reduced speckle and improved visibility of the underlying chart features. As expected from the blur law, the averaged images exhibit reduced spatial resolution, and additional scattering from milk and tissue attenuates the overall signal, reducing the dynamic range of the *en face* images relative to OCT-only.

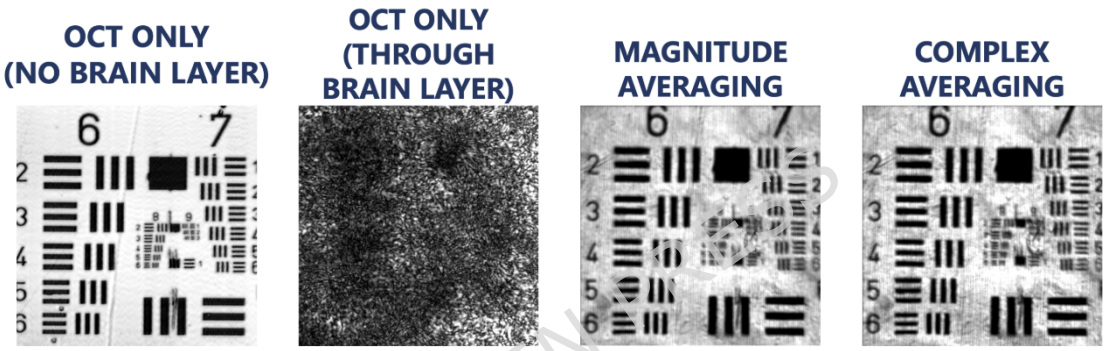


Fig. 7. Imaging a USAF resolution chart embedded in milk (DSM) and covered by a 250 µm-thick mouse brain slice using DSM-enhanced OCT with N=100 realizations.

The simulations (Figs. 2–4) reproduce the main experimental trends. Both averaging modes blur edges according to the same σ_θ, while complex averaging maintains higher contrast by eliminating the additive pedestal. The static-speckle simulations (Fig. 4) model the tape layer and the brain slice as static phase screens of different strength and grain size. The experiments reproduce the same pattern. Specifically, granular distortions vanish after averaging, the complex background is darker due to reduced coherent throughput, and blur remains governed by σ_θ.

Discussion

Our results show that a dynamic scattering medium (DSM) can serve as an in-situ coherence modulator for coherent imaging. The time-dependent pupil model, which includes small random tilts plus higher-order dynamic phase, predicts two universal outcomes that we observe throughout. First, the speckle and low-order distortions are suppressed when averaging independent views. Second, the effective PSF broadens by a small, predictable kernel set by the tilt variance σ_θ^2 (the blur law). This separation of roles appears clearly in the simulations. At fixed σ_θ the edge width Δx remains essentially unchanged when we vary the dynamic phase depth σ_{dyn}, whereas the Michelson contrast and speckle contrast improve as σ_{dyn} decreases. The experiments reproduce the same pattern. In the DSM-averaged OCT data, both magnitude and complex images exhibit modest edge broadening relative to OCT-only, and complex averaging consistently delivers higher contrast because it removes the additive background (dynamic pedestal) in the mean.

The averaging strategy used here has a direct conceptual analogy to classical atmospheric turbulence mitigation [45, 46], which acquires many realizations of a static scene under independent turbulent wavefronts, registers frame-to-frame tilts, temporally averages the registered frames, and finally deconvolves the averaged image against a (typically Gaussian) blur model. Our DSM-enhanced OCT pipeline implements the averaging step and, uniquely, because the signal is complex, enables field (complex) averaging that additionally suppresses the incoherent pedestal that intensity-only turbulence methods cannot remove. Because Eqs. (6)-(7) provide the analytic blur kernel, a principled space-invariant deconvolution can be applied on top of averaging to further reduce the residual blur. We identify per-realization tilt registration and predictive-kernel deconvolution as natural future extensions.

The simulations with a thin static phase screen connect directly to the tape and brain-slice experiments. In both,

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the static phase creates granular distortions in a single frame; DSM-induced angular diversity decorrelates those patterns across views, so averaging reduces them to a low-frequency halo in magnitude images and to a darker coherent background in complex images. When the static phase is moderate ($\sigma_{stat} \lesssim 1$ rad), complex averaging recovers the object cleanly with lower residual speckle. When the screen is strong ($\sigma_{stat} \gg 1$ rad), the coherent throughput $m_{stat} \approx e^{-\sigma_{stat}^2/2}$ becomes small and the complex image is darker, which is an expected limit rather than a failure of the mechanism.

Because speckle contrast is often reported as $K = \sigma/\langle I \rangle$, it is important to interpret K alongside the local mean intensity in the ROI. In strong-screen cases the darker complex background can inflate K numerically even when the additive pedestal is absent.

The model separates control parameters for resolution and contrast. Resolution is governed by σ_θ via the blur law, while contrast is governed by σ_{dyn} (and N) through the pedestal and residual speckle. For a target Δx, choose σ_θ (via the blur law). Choose σ_{dyn} and N to meet a target speckle level (e.g., using K), and set the inter-volume delay Δt above the DSM decorrelation time to ensure statistical independence (e.g., $\Delta t = \lambda^2/(16D)$ under Brownian diffusion). These rules let one tune the trade-off a priori.

Methodologically, complex averaging requires phase stabilization. Our data show that even simple global alignment works in practice, but excessive spatial smoothing of the phase can erode the advantage of complex over magnitude averaging. Conversely, magnitude averaging is robust and simple, but it always retains the additive offset that lowers contrast. In both modes the edge blur is the same, because σ_θ is the same.

The method is not an aberration-correction technique. It reduces speckle and coherent artefacts by averaging over independent DSM realizations, but cannot recover spatial frequencies lost when the aberrated PSF core is already severely broadened. Quantitatively, when the wavefront aberration RMS exceeds approximately 1 rad at the pupil, so that the coherent peak of the PSF is destroyed, averaging does not help. This is an intrinsic upper bound that applies to any averaging-based scheme. In the regime where a detectable coherent peak survives, our simulations (see Figs. S2 and S3) show that performance depends only weakly on the Zernike order of the aberration, as expected from the blur law. Finally, blood (in contrast to milk) is strongly absorbing and biologically variable, and would additionally introduce flow-dependent decorrelation; we therefore perfused animals with saline to isolate the DSM effect cleanly.

Although we demonstrate the approach with OCT, the mechanism is modality-agnostic. Any coherent imaging system that can form repeated, independent views (e.g., digital holography, full-field coherent microscopy) can use a DSM to reduce effective spatial coherence without SLMs or complex wavefront control. The concept can be implemented as a DSM-immersed objective or as a compact add-on coherence-control cell (e.g., a microfluidic reservoir with controlled flow) that allows control of σ_θ, σ_{dyn}, and Δt. Such modules could deliver speckle-reduced, aberration-robust imaging with a small and predictable resolution penalty, set at design time.

This study focuses on static samples. *In vivo* use will require faster acquisition or DSMs with shorter decorrelation times. We did not map the full (σ_θ,σ_{dyn},σ_{stat},l_{stat}) space experimentally. Future work will include B-scan analysis through tissue, controls against water-immersion and rotating diffusers and tunable DSM prototypes that validate the design rules prospectively.

Conclusions

A dynamic scattering medium (DSM) can act as an in-situ spatial-coherence modulator for coherent imaging. Averaging angularly and phase-diverse views reduces the effective spatial coherence, suppressing speckle and low-order distortions while introducing a predictable blur described by the blur law. Simulations and experiments show that the blur is set primarily by the angular diversity σ_θ. After phase stabilization, complex averaging suppresses the additive pedestal retained by magnitude averaging, yielding higher Michelson contrast.

Static phase structure adds one practical constraint. The strength of a static screen limits coherent throughput, and its correlation length l_{stat} shapes how the background looks after averaging: fine screens produce higher-frequency residuals, whereas moderate–coarse screens yield smoother backgrounds and cleaner complex reconstructions. In practice, when using complex averaging it helps to avoid very strong static screens and, where mounting permits, bias the preparation toward a coarser effective l_{stat} (e.g., by smoothing interfaces or light index matching), the residual blur remains governed by angular diversity.

Although *in vivo* use may require faster decorrelation, the method is directly applicable to *ex vivo* and static preparations and is compatible with water-immersion objectives. It can be implemented in DSM-immersed microscopes or compact coherence-control modules to reduce speckle and aberration sensitivity while introducing a quantifiable blur set by the design parameters. The approach is transferable beyond OCT to other coherent imaging modalities.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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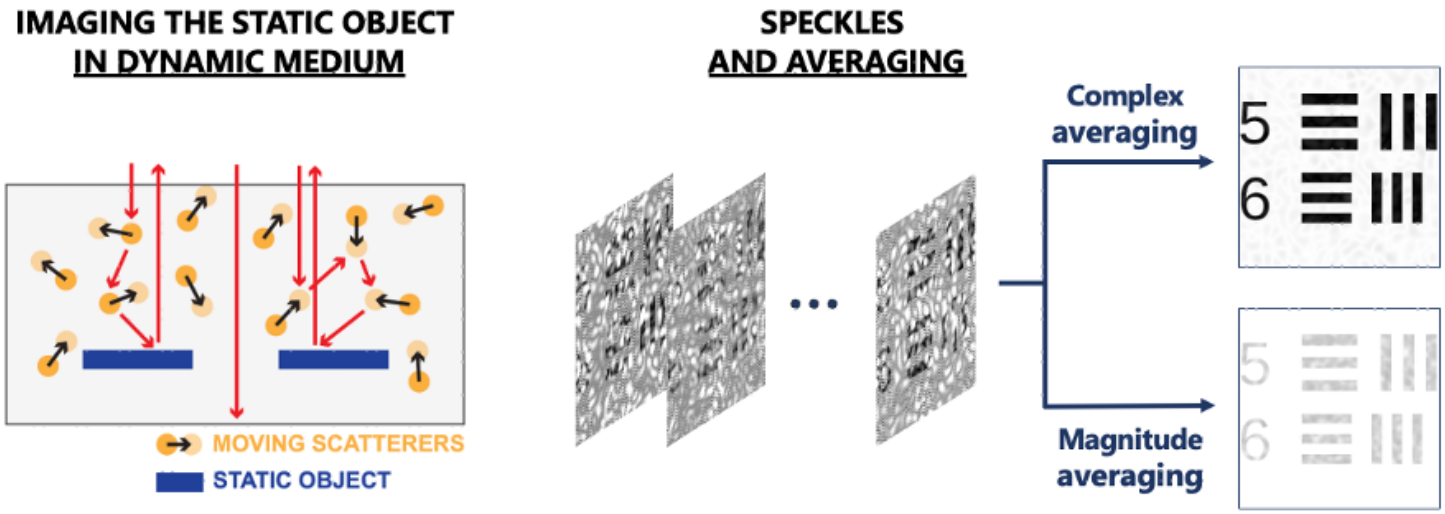
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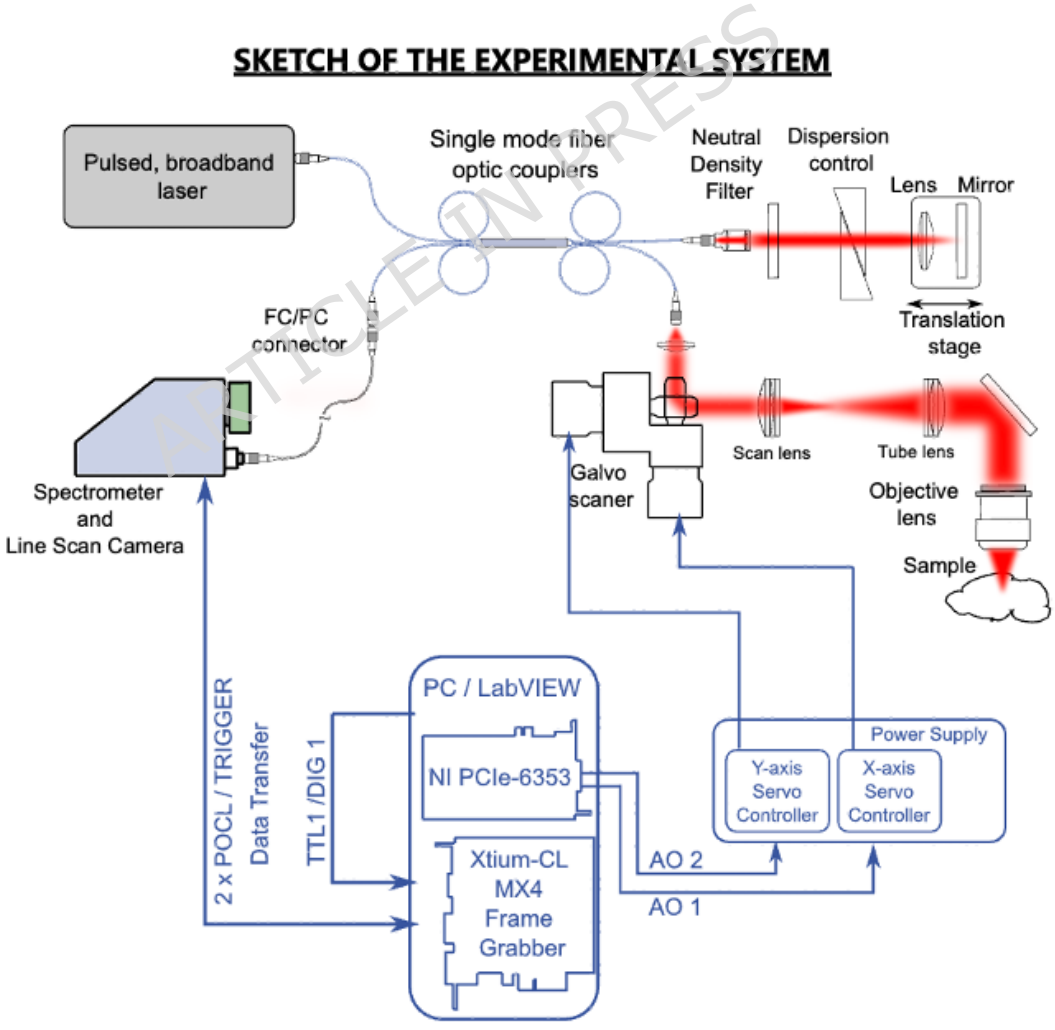
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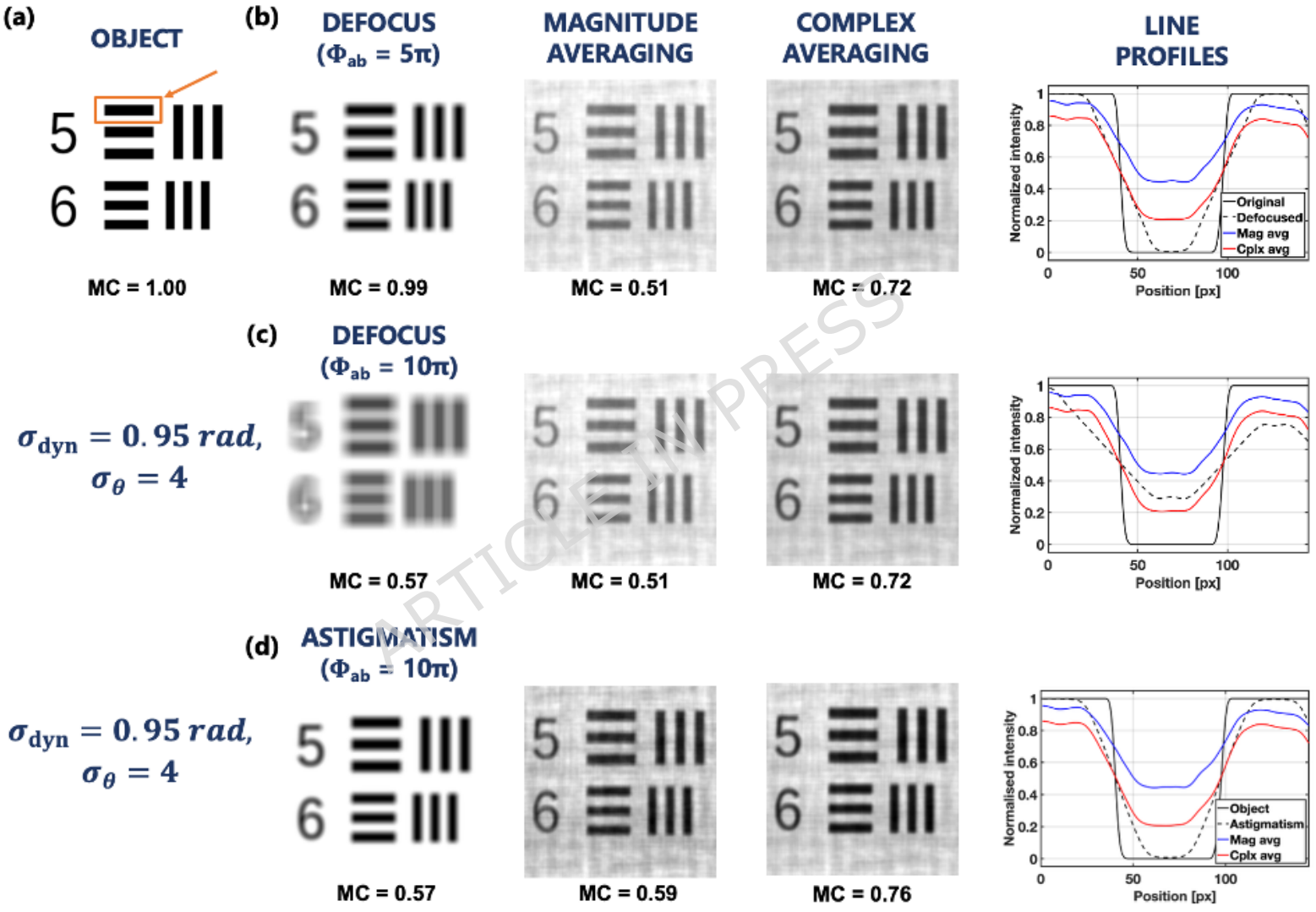
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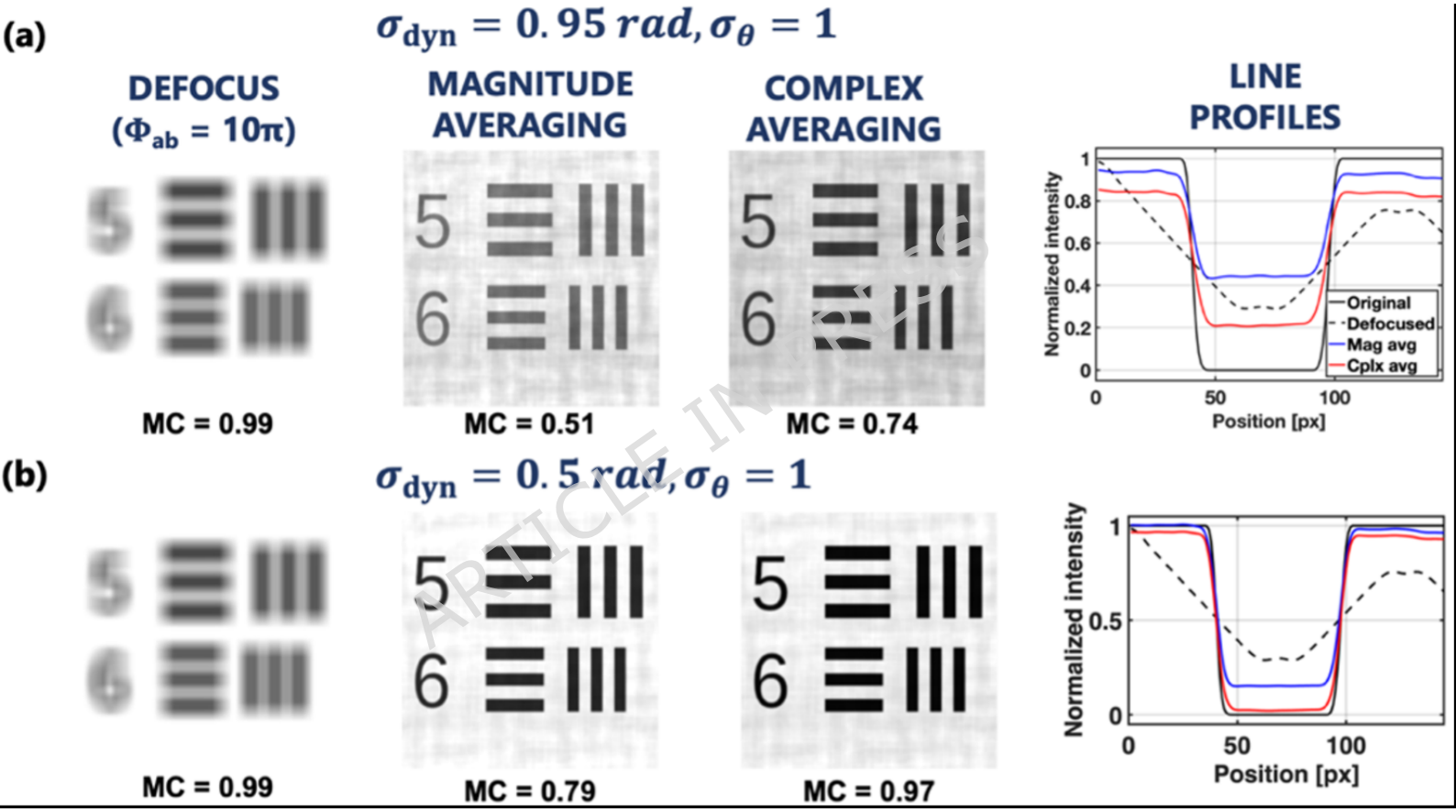
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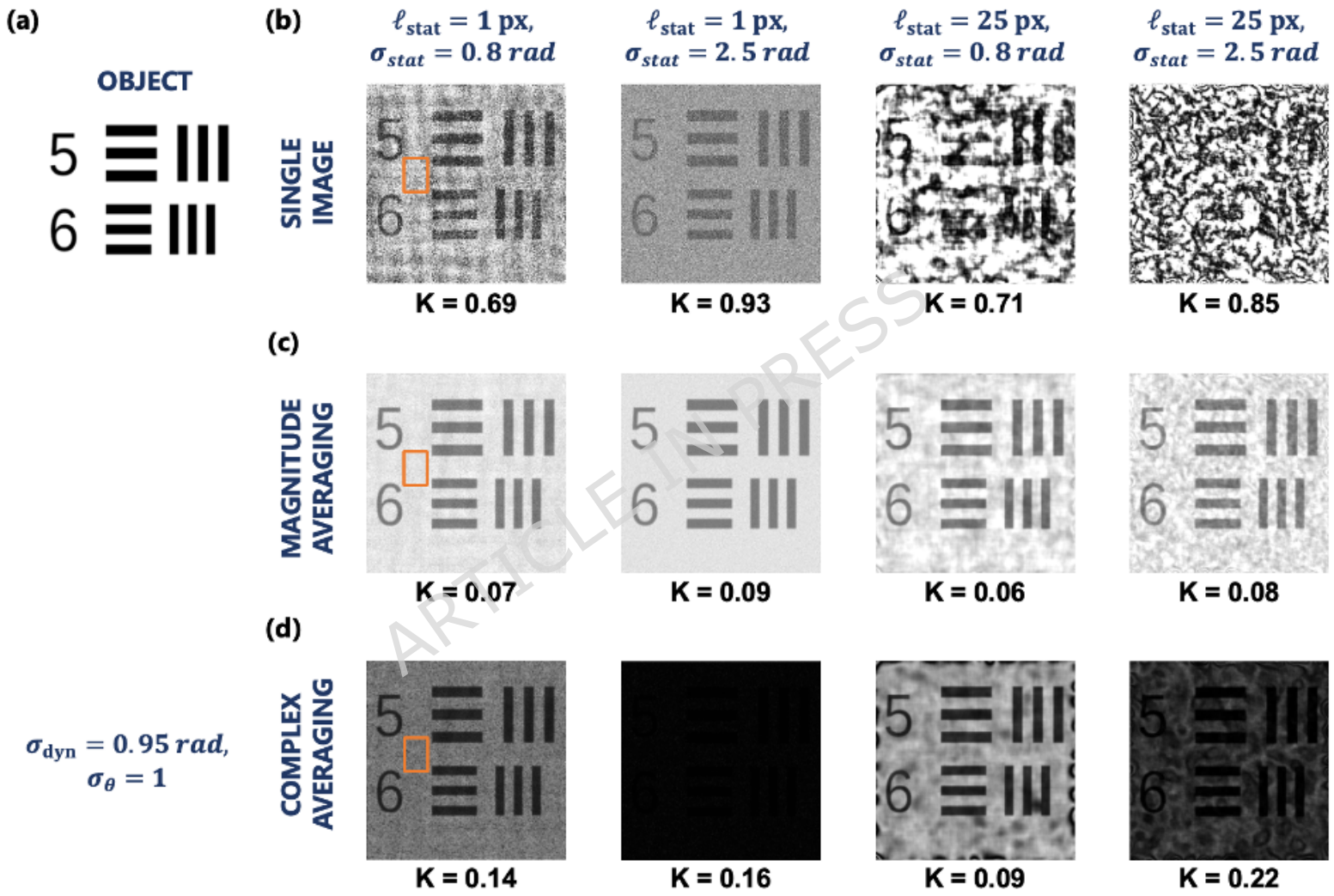




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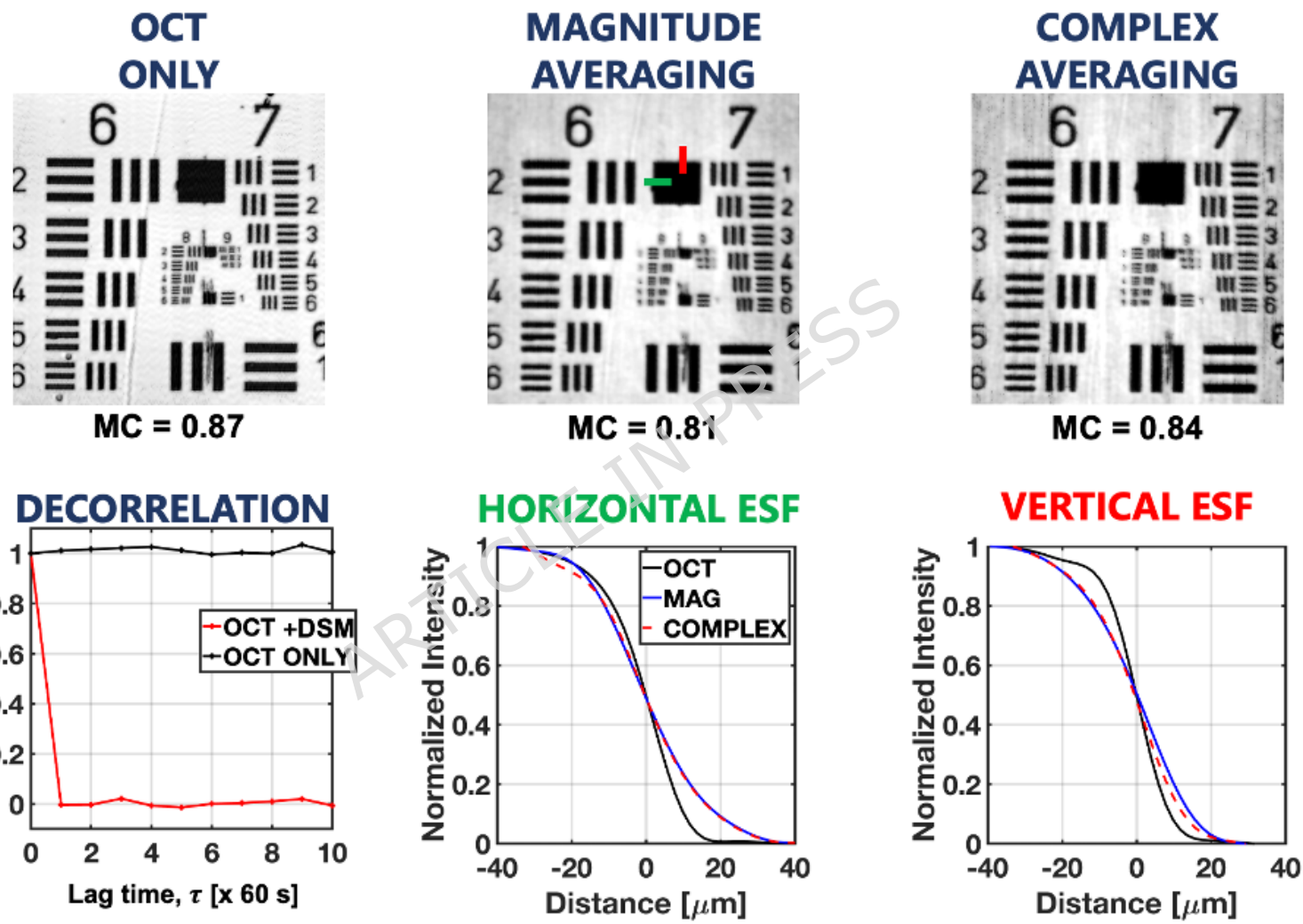


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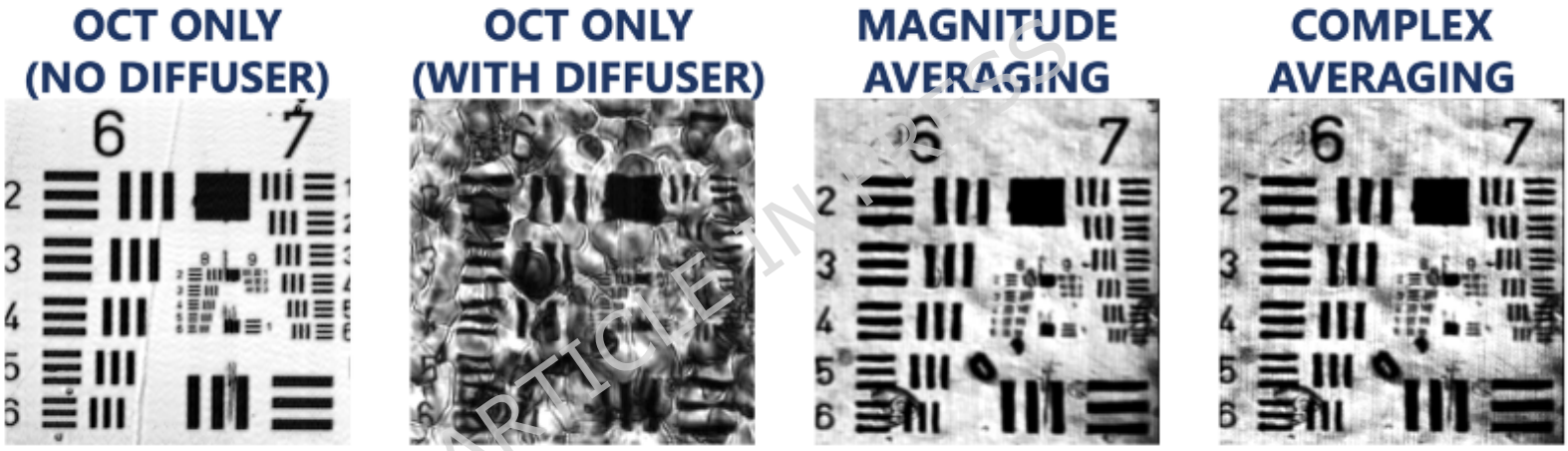


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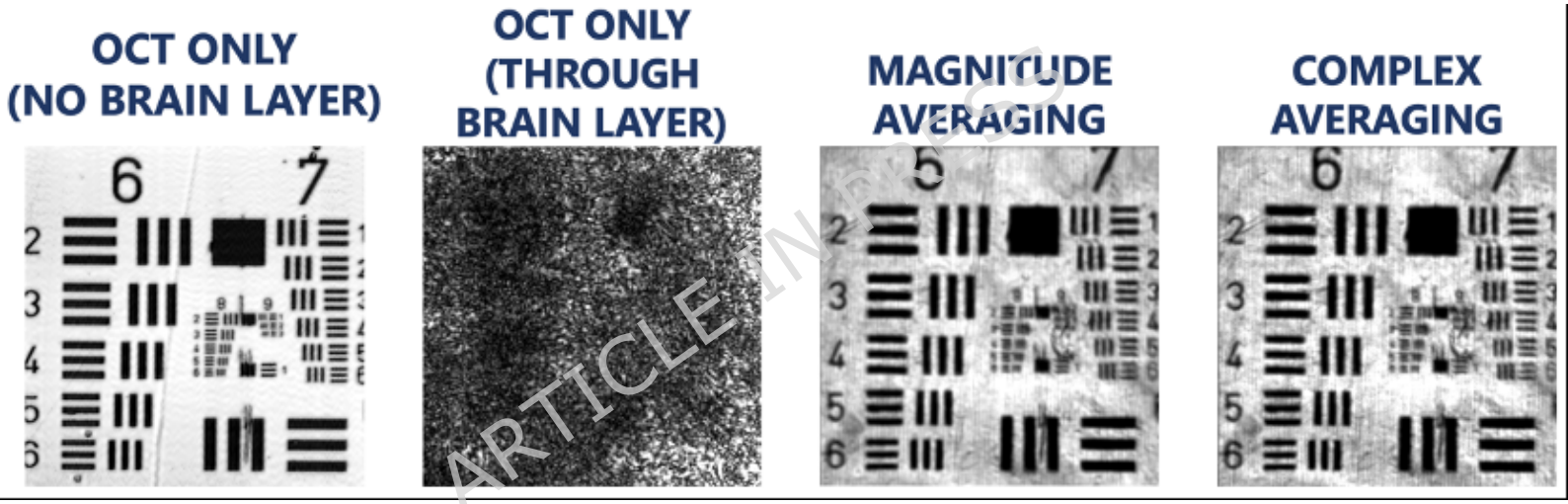


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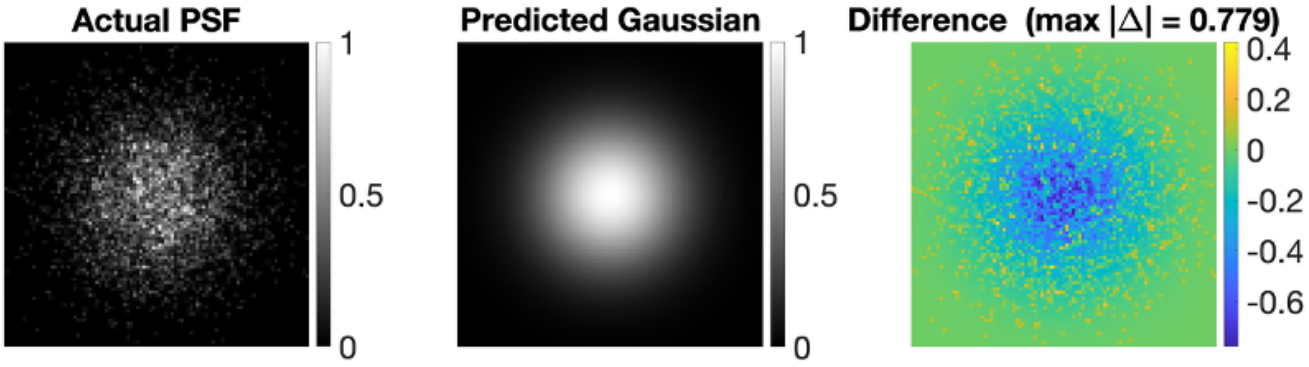
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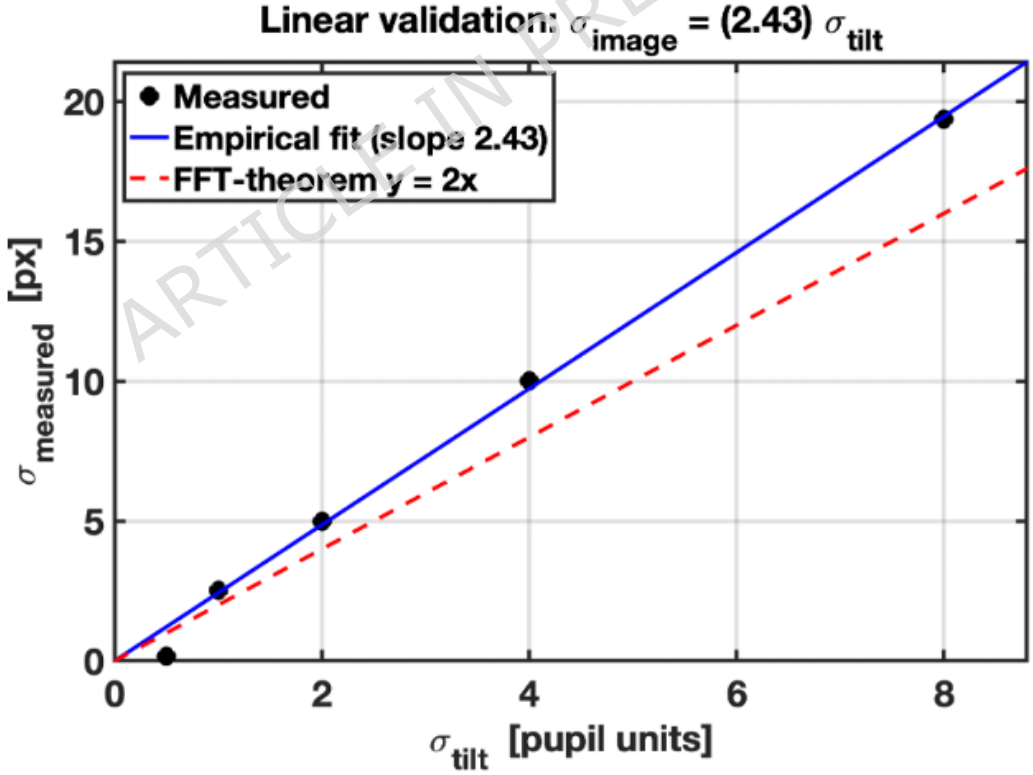
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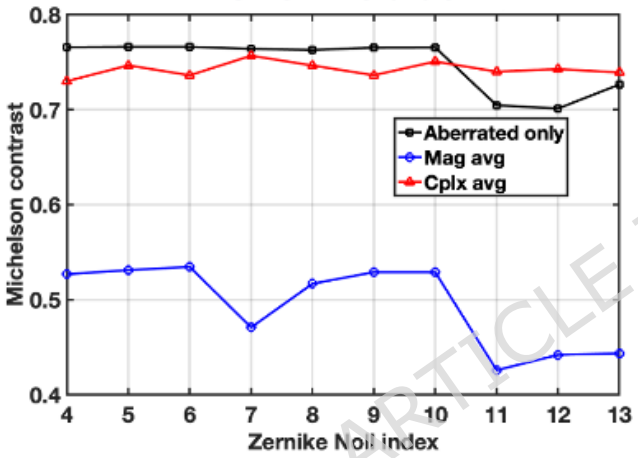
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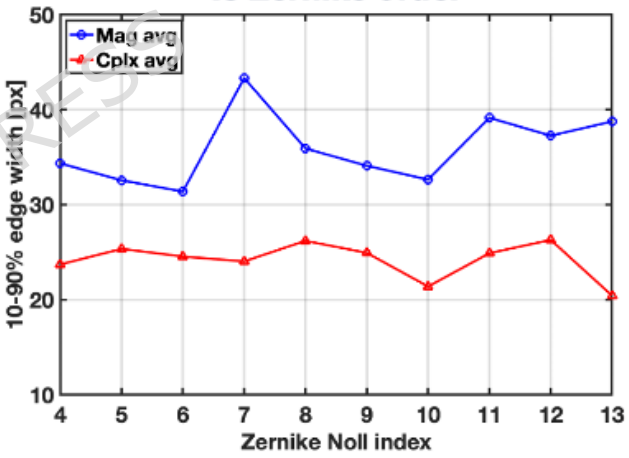
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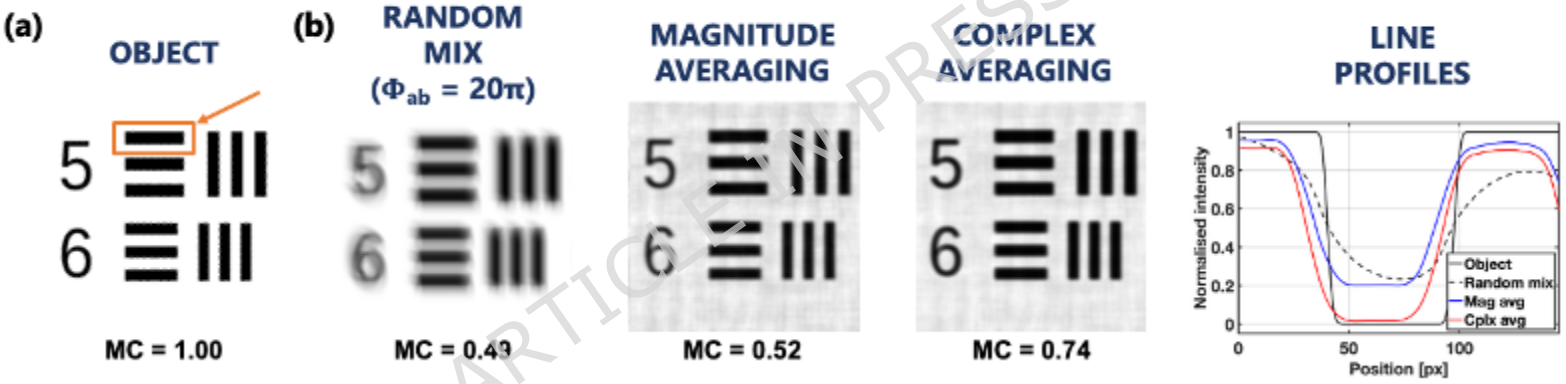
(b)



$\sigma_{\text{dyn}} = 0.5\,rad, \sigma_\theta = 4, \Phi_{ab} = 5\pi, N = 100$

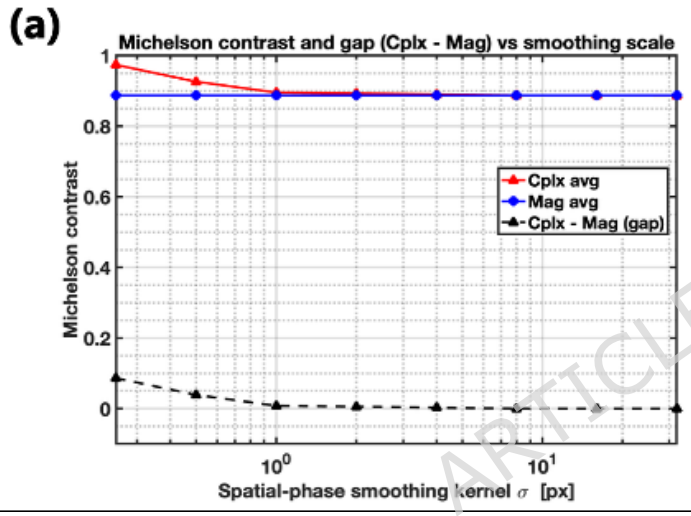
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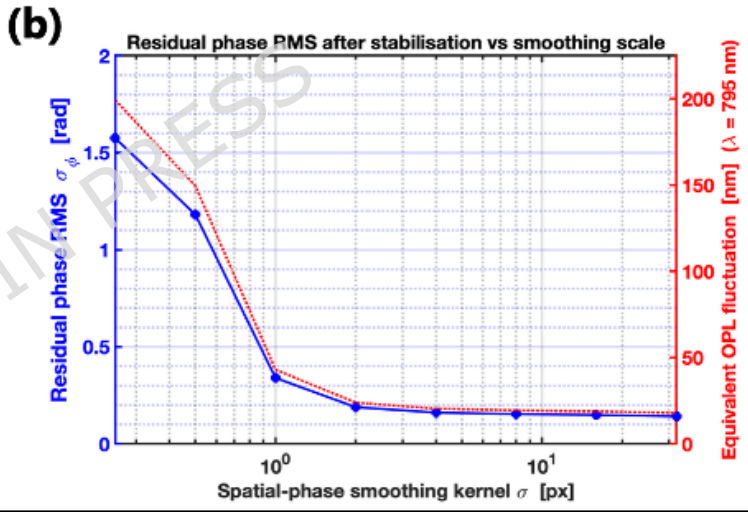
$$\sigma_{\mathrm{dyn}} = 0.5\,rad, \sigma_{\theta} = 4, N = 100$$



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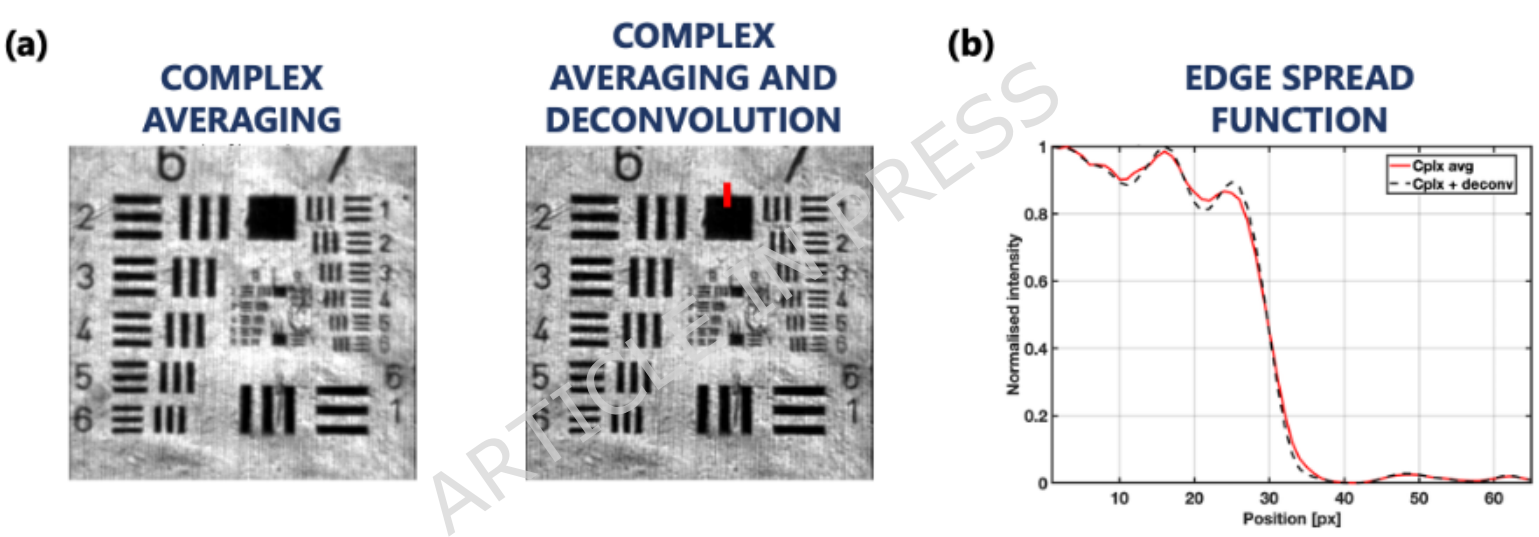
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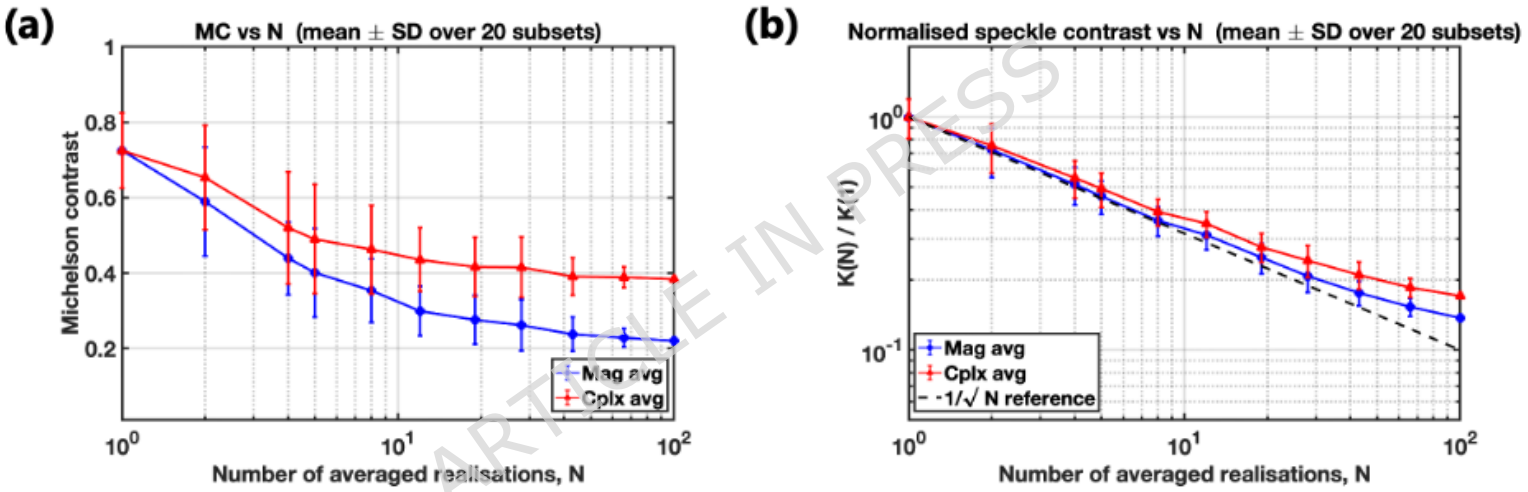
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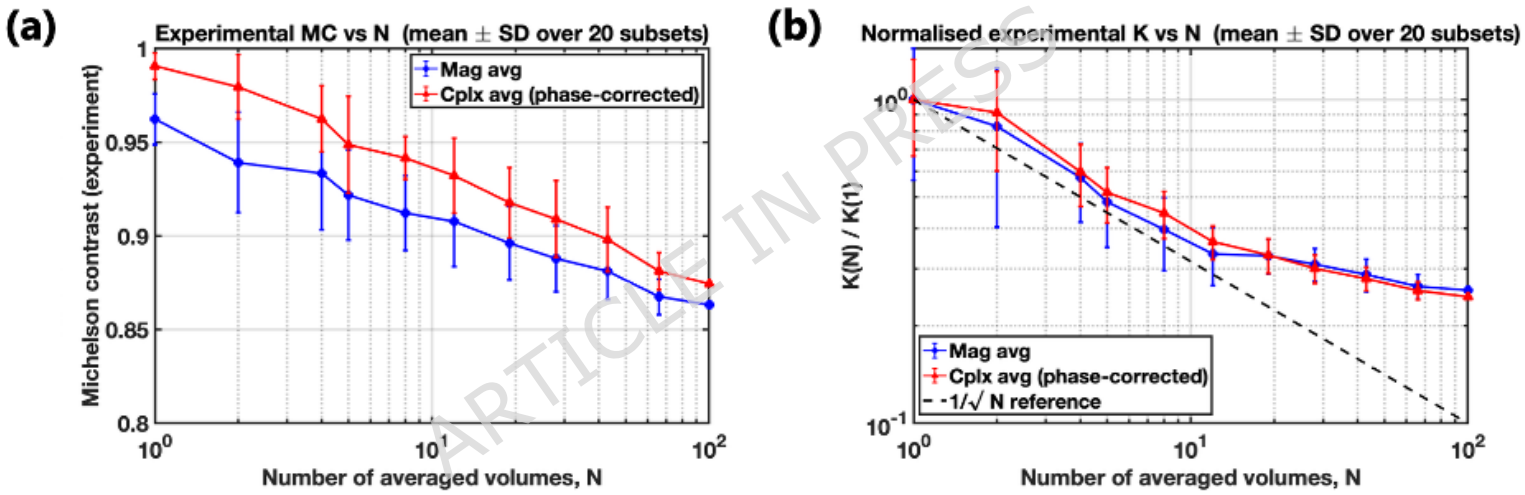
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BRAIN LAYER



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